

Command-response modeling of spontaneous Swiss German dialectal speech*

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1 Introduction

This study presents preliminary results of an intonational modelling of spontaneous Swiss German (SG) dialectal speech by the application of the command response model [1]. 40 Speakers from 4 dialect regions in Switzerland - 2 Alpine and 2 Midland dialects - were analyzed as follows: broad transcription of the data, segmentation in *Praat* [2], and annotation with variables that are assumed to occupy a crucial role for a cross-dialectal intonational description. The data was parametrized by means of Mixdorff's *FujiParaEditor* [3]. The analysis is based on approximately 3 minutes of spontaneous speech per speaker, which amounts to 41'000 syllables total. In what follows, preliminary results are presented.

2 Tested Variables

On the global intonation level, it is primarily the phrase command magnitude (Ap) as well as intonation phrase (IP) length which are of interest. Local descriptors involve accent command amplitude (Aa) as well as the timing and length of the accent command. Together with the labeled variables word class, stress, and syllable nucleus type, these factors constitute the main lines along which dialectal variation of intonation proper is studied. Yet, Hirst [4], for instance, points out that “an adequate description of intonation needs to take into account not simply the phonology of the language, but also the syntax and the semantics, as well as the interfaces between the grammar and the “real world” constituted by phonetics and pragmatics” (93). Thus, this study has the – admittedly ambitious – demand of approaching intonation holistically, that is,

conversational and information-structuring functions of intonation as well as the paralinguistic component of *emotion* are, too, taken into consideration. These factors shall also undergo testing for inter-dialectal variation by means of the command-response model [1]. Note that some of these results are not in line with previous results of the project, e.g. [5] – reflecting a modified, enhanced data analysis and parametrization method.

3 Preliminary Results

3.1 Phrase component

Figure 1 shows Ap magnitudes according to the 4 dialect groups Bern BE, Grisons GR, Valais WS, and Zürich ZH.

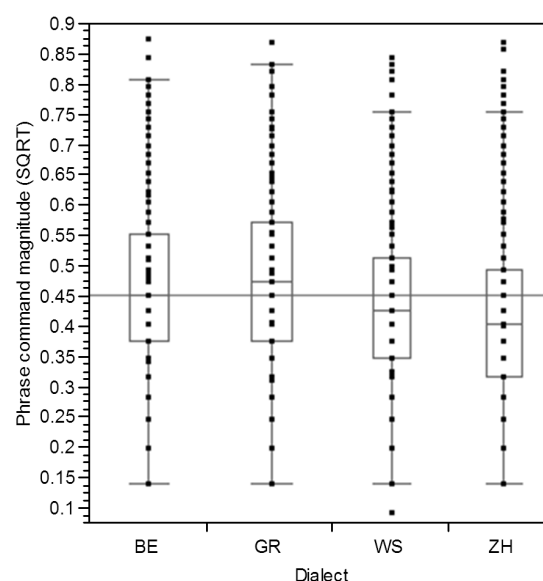


Fig. 1 Ap of the four dialects.

The dialects are all significantly different from each other in their Ap values, the south eastern Alpine variety GR shows the highest values – the Midland Eastern variety ZH the lowest. As for phrase magnitude and emotion: the BE

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distinguish between high magnitude neutral speech and low magnitude bored speech – while the WS show high magnitude bored speech in contrast to low magnitude happy and fearful speech.

3.2 Accent component

Table 1 shows tentative results of the observed accent command variables in the context of the 4 dialects.

Table 1 Tentative results of the accent component, according to amplitude and timing.

Overall amplitude	Alpine speakers (WS and GR) show significantly higher amplitudes than the Midland speakers, particularly BE.
Stress and amplitude	All dialects show higher amplitudes for metrically stressed syllables.
Word class and amplitude	All dialects indicate higher amplitudes for syllables belonging to lexical - as opposed to grammatical - words.
Syllable nucleus and amplitude	On the whole, long vowel syllable nuclei show significantly higher amplitudes than diphthong, schwa, and short vowel nuclei.
Phrase type and amplitude	All dialects demonstrate higher amplitudes in continuing phrases as opposed to in terminating phrases.
Emotion and amplitude	The WS speakers indicate higher amplitudes in happy and neutral states as opposed to in bored states. The BE show peak amplitudes in happy states while lowest amplitudes are found in bored speech and disgust states. The ZH, on the contrary, mark disgust speech with high amplitudes, contrasting low-amplitude bored and fearful speech.
Focus and amplitude	In all dialects, focused words are marked with higher amplitudes than words which are not in focus.
Timing	BE speakers have late accent command onsets, only 2 msec before the syllable beginning. GR shows the earliest onsets, on average 31 msec before the syllable; the WS and ZH are positioned between the two extremes.

The results of accent timing and amplitude point to significant differences but also reveal striking similarities in the local accentual structuring of the observed dialects. The similarities – higher amplitudes in stressed syllables, lexical words as well as in focused constituents - confirm what is frequently presented in the literature. Differences which

need to be pursued in more detail are particularly Ap and Aa magnitude and amplitude. On the global level, correlations between Ap and pre-IP pause length, as well as between Ap magnitude and the length of the previous IP can be anticipated. Locally, accent command positioning in the IP – among other things - needs to be specified and accounted for.

4 Conclusions

The different weight of the observed variables renders conclusions increasingly difficult – which in turn makes the object under observation even more intriguing. This ongoing work aspires, in the end, to establish a generalized linear model which will shed light on the weight of the individual factors that cause inter-dialectal intonational variation. Thereby, this work partially fills a research gap on SG dialectal intonation, in doing so, it hopes to provide fertile grounds for future research.

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